NETHRA CHANDRAGANDHI T

240701357

EXP:08 – WORKFLOW USING ZOHO CREATOR

1. Aim

To create and test workflows using Deluge scripts and email actions in Zoho Creator for the Internship Management System, automating attendance percentage, performance result, working hours, approval notifications, internship duration, and application approval.

2. Procedure

1. Open the Internship Management System in Zoho Creator and select the Workflow tab.
2. Choose the form and create a workflow for the required record event or field update.
3. Add a Deluge script or Send Mail action according to the workflow requirement.
4. Save and enable the workflow.
5. Enter or edit sample records to trigger the workflow.
6. Verify the calculated values, updated status fields, or email action and capture the output.

3. Workflows Implemented

3.1 Calculate Attendance Percentage

Purpose: Calculates a student's attendance percentage from attendance records. Present days count as one attended day, Absent days count only toward total days, and Half Day counts as 0.5 attended day.

Logic: The script fetches the student's attendance records, totals the relevant attendance days, calculates (attended days / total days) × 100, and stores the result in Attendance Percentage. If there are no applicable records, the percentage is set to 0.

```
attendance_records = Internship_Attendance[Student == input.ID];
total_days = 0;
attended_days = 0.0;
for each  record in attendance_records
{
    if(record.Attendance_Status == "Present")
    {
        total_days = total_days + 1;
        attended_days = attended_days + 1;
    }
    else if(record.Attendance_Status == "Absent")
    {
        total_days = total_days + 1;
    }
    else if(record.Attendance_Status == "Half Day")
    {
        total_days = total_days + 1;
        attended_days = attended_days + 0.5;
    }
}
if(total_days > 0)
{
    input.Attendance_Percentage = (attended_days / total_days * 100).round(2);
}
else
{
    input.Attendance_Percentage = 0;
}
```

Figure 3.1: Calculate Attendance Percentage Deluge script.

3.2 Calculate Performance Result

Purpose: Assigns a performance label based on Overall Rating.

Logic: Overall Rating ≥ 4.5 gives Excellent; ≥ 3.5 gives Good; ≥ 2.5 gives Average; otherwise, the result is Poor. The selected label is stored in Performance Result.

```
if(input.Overall_Rating >= 4.5)
{
    input.Performance_Result = "Excellent";
}
else if(input.Overall_Rating >= 3.5)
{
    input.Performance_Result = "Good";
}
else if(input.Overall_Rating >= 2.5)
{
    input.Performance_Result = "Average";
}
else
{
    input.Performance_Result = "Poor";
}

```

Figure 3.2: Calculate Performance Result Deluge script.

3.3 Attendance Working Hours Workflow

Purpose: Calculates the number of working hours from Check-In Time and Check-Out Time.

Logic: Converts both times into seconds, finds the difference, divides by 3600, and rounds the value to two decimal places. If the check-out time is earlier than the check-in time, Working Hours is set to 0.

Test result: Check-in at 09:00 and check-out at 15:00 produces 6.00 working hours.

```
if(input.Check_In_Time != null && input.Check_Out_Time != null)
{
    checkin_seconds = input.Check_In_Time.getHour() * 3600 + input.Check_In_Time.getMinutes() * 60 + input.Check_In_Time.getSeconds();
    checkout_seconds = input.Check_Out_Time.getHour() * 3600 + input.Check_Out_Time.getMinutes() * 60 + input.Check_Out_Time.getSeconds();
    if(checkout_seconds >= checkin_seconds)
    {
        input.Working_Hours = ((checkout_seconds - checkin_seconds) / 3600.0).round(2);
    }
    else
    {
        input.Working_Hours = 0;
    }
}
```

Figure 3.3: Attendance Working Hours Deluge script.

3.4 Student Approval Email Notification

Purpose: Sends an email message when an internship application is approved.

The Send Mail action contains the sender, recipient, subject “Internship Application Approved,” and a message asking the student to log in to view the application details. The screenshot shows the configured test recipient; for deployment, the recipient should be mapped to the relevant student's email field if notifications must be individualized.

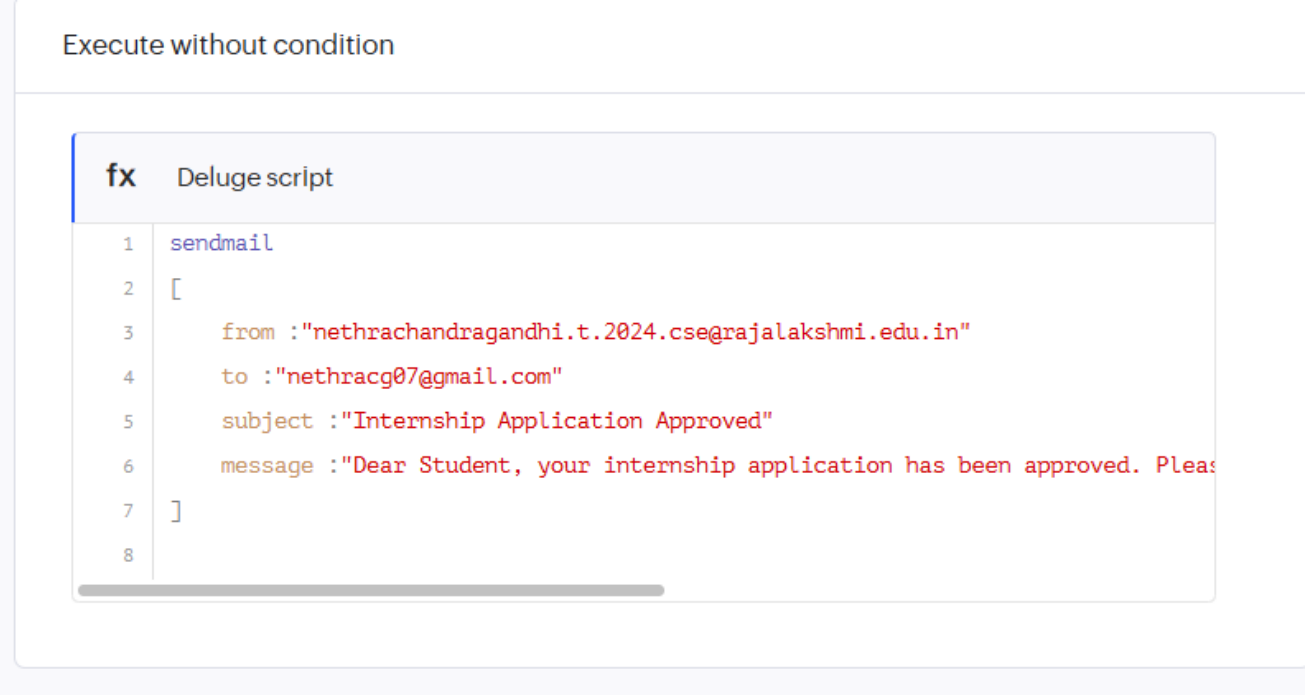


Figure 3.4: Student Approval Email Notification Send Mail action.

3.5 Calculate Internship Duration

Purpose: Calculates the number of days between Start Date and End Date and stores the value in Internship Duration.

Logic: The calculation runs only when both dates are present. For 01-Nov-2026 to 01-Dec-2026, the expected duration is 30 days.

Execute without condition

fx Deluge script

```
1  if(input.Start_Date != null && input.End_Date != null)
2  {
3      duration = input.Start_Date.daysBetween(input.End_Date);
4      input.Internship_Duration = duration;
5  }
6
```

Figure 3.5: Calculate Internship Duration Deluge script.

3.6 Application Approval Workflow

Purpose: Updates application status fields when Application Approved is selected.

Logic: If Application Approved is true, Approval Status is set to Approved and Application Status is set to Selected.

Execute without condition

fx Deluge script

```
1  if(input.Application_Approved == true)
2  {
3      input.Approval_Status = "Approved";
4      input.Application_Status = "Selected";
5  }
6
```

Figure 3.6: Application Approval Workflow Deluge script.

4. Result

The six workflow scripts/actions used in the Internship Management System were documented with their actual Zoho Creator screenshots. They automate attendance percentage calculation, performance classification, attendance working-hour calculation, approval email notification, internship duration calculation, and application approval status updates.

5. Conclusion

Zoho Creator workflows and Deluge scripts reduce manual calculations and repetitive updates in the Internship Management System. The implemented logic helps maintain consistent attendance, evaluation, internship-duration, and application-approval information.